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The Application of Logistic Regression to Examine the Predictors of Alcohol-Involved Crashes

1. Introduction

Drunk driving is perhaps one of the most terrifying and preventable realities that many of us must deal with every day. According to the U.S. Department of Transportation, one American is killed in a car crash by a driver under the influence of alcohol every 51 minutes. According to the latest WHO data (from 2017), the U.S. has one of the highest proportions of road traffic deaths attributed to alcohol in the developed world.¹ At 29%, it is tied for third with France and Portugal, trailing Canada (29.6%) and Ireland (38.5%). These numbers are very high, especially when compared to countries like Mexico (19.5%), Australia (17%) and Germany (7%).

Recognizing that this problem persists across the nation, in this analysis we focus specifically on Philadelphia. Pennsylvania ranks 10th in drunk driving crash deaths behind Illinois and ahead of New York according to the National Highway Traffic Safety Administration and, thus, serves as a key candidate for any interventions to prevent these deaths.²

In this paper, we apply logistic regression to examine predictors of alcohol-involved crashes using Census and Pennsylvania Department of Transportation geocoded crash data at the block group level. We estimate a logistic regression model in which the binary dependent variable is `DRINKING_D`, indicating whether a crash involved a drinking driver. The model includes the independent variables `FATAL_OR_M` (if the crash involved a fatality or major injury), `OVERTURNED` (whether the vehicle was overturned in the crash), `CELL_PHONE` (whether a driver was using their phone), `SPEEDING` (whether a driver was speeding), `AGGRESSIVE` (whether a driver was driving aggressively), `DRIVER1617` (whether a driver was 16-17 years old), `DRIVER65PLUS` (whether a driver was over 65 years old), `PCTBACHMOR` (percentage of people over 25 years old with a bachelor's degree of the census block group where a crash occurred), and `MEDHHINC` (median income of the census block group where a crash occurred) using R. Several of these variables show promise as predictors of the odds a crash involved a drunk driver. Particularly, we speculate that `DRIVER1617` would have a negative impact on drunk driver odds given difficulty obtaining large quantities of alcohol (and in many cases a vehicle at all) at this young age. We also hypothesize that median income and bachelor's degree attainment will have an inverse impact on these odds (i.e. crashes occurring in block groups with lower income and lower educational attainment may have different odds of alcohol

¹ [https://www.who.int/data/gho/data/indicators/indicator-details/GHO/attribution-of-road-traffic-deaths-to-alcohol-\(-\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/attribution-of-road-traffic-deaths-to-alcohol-(-))

² <https://www-fars.nhtsa.dot.gov/States/StatesAlcohol.aspx>

involvement), given the potentially disproportionate presence of car-centric neighborhoods and establishments selling more inexpensive alcohol in these block groups.

2. Methods

2.1 Logistic Regression

In OLS, coefficients are interpreted as the amount by which the dependent variable (DV) changes when the independent variable (IV) increases by 1 unit. When the dependent variable is binary, OLS is not ideal because the fitted values are interpreted as probabilities but are not constrained to fall between 0 and 1. For instance, a coefficient for an IV in an OLS regression could be 0.5, meaning the DV increases by 0.5 per 1 unit increase in the IV. When the dependent variable is binary, OLS is not ideal because the fitted values are interpreted as probabilities but are not constrained to fall between 0 and 1. OLS also assumes a linear relationship between the predictors and the outcome, which implies constant marginal effects across the full range of predicted probabilities. In addition, the error variance is not constant for a binary outcome, so the homoskedasticity assumption is usually violated. For these reasons, logistic regression is more appropriate for modeling the probability that $DRINKING_D = 1$.

Logistic regression uses a workaround that allows the dependent variable to be binary. First, one must understand odds and the odds ratio. Probability is calculated as:

$$P(Y = 1) = \# \text{ desirable outcomes} / \# \text{ possible outcomes}$$

Odds are calculated as:

$$Odds(Y = 1) = \# \text{ desirable outcomes} / \# \text{ undesirable outcomes}$$

Or

$$Odds = \frac{p}{1 - p}$$

Where p = probability

In simple terms, probability is the ratio that a desired outcome occurs to all possible outcomes, and odds is the ratio of desirable outcomes occurring to undesirable outcomes. For example, if there is a 0.67 (two in three) probability of something happening, the *odds* are $0.67/0.33 = 2$, meaning the event is two times more likely to occur than not.

To get around the issue of predicting the value of a binary Y, logistic regression instead predicts the probability that the DV = 1 given the IV's value. In other terms, logistic regression predicts $P(Y = 1 | X = x)$. For instance, when predicting whether or not a zip code will have a hospital in

it ($Y = 1$ or 0), an increase in population of that zipcode by 1000 people might increase the odds that there is a hospital (that $Y = 1$). Whereas linear regression can predict values from $-\infty$ to $+\infty$, probability can only range between 0 and 1, thus linear regression can produce fitted probabilities below 0 or above 1, which does not make sense for a probability. To get around this we need a function that translates coefficients to a format that makes sense for a binary DV. One function is the Logit Function:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \varepsilon$$

Where $p = P(Y = 1)$ and $p / 1 - p$ is the odds

Log odds (or logit) = the natural log of $(p / (1 - p))$

β_0 = the intercept (in our case, the log-odds of DRINKING_D = 1 when all predictors are zero

β = the slope coefficient for each predictor

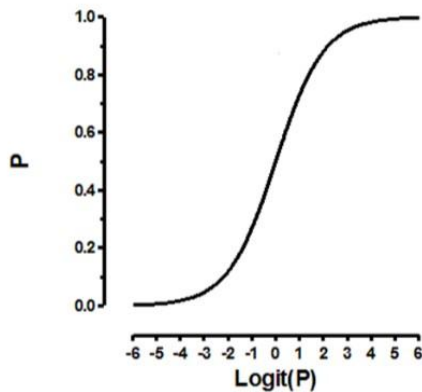
For ease of comprehension and to keep the equation short and legible in this document, we just used the placeholder x . To transform the equation so that it is in the " $P(Y = 1) =$ " form, we simply exponentiate the right side of the equation. Here is the *logistic* form of the equation, this time with our model's variables:

$$P(\text{DRINKING_D} = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \text{FATAL_OR_M} + \beta_2 \text{OVERTURNED} + \beta_3 \text{CELL_PHONE} + \beta_4 \text{SPEEDING} + \beta_5 \text{AGGRESSIVE} + \text{DRIVER1617}\beta_6 + \text{DRIVER65PLUS}\beta_7 + \beta_8 \text{PCTBACHMOR} + \text{MEDHHINC}\beta_9)}}$$

Where e = the Euler's number (~ 2.718), the base of the natural log ($\ln$)

The logistic function works for binary outcomes because it is bounded between 0 and 1. Since the exponentiated variable is on the denominator of the equation, $P(\text{DRINKING_D})$ can never be greater than 1, and since the denominator is $1 + e^{\text{the variables and their multipliers}}$, the denominator can never be negative. Thus, p will never be less than 0 or greater than 1.

Furthermore, the logistic function has a non-linear, S-shaped curve that makes it progressively more difficult to increase or decrease the probability of an event happening the closer it gets to $p = 1$ or $p = 0$, respectively.



2.2 Hypothesis Testing

For each predictor, FATAL_OR_M, OVERTURNED, CELL_PHONE, SPEEDING, AGGRESSIVE, DRIVER1617, DRIVER65PLUS, PCTBACHMOR, and MEDHHINC, the null hypothesis is that the coefficient equals zero: $H_0: \beta_i = 0$. This means the predictor is not associated with the log-odds that a crash involved a drinking driver, holding all other variables constant. The alternative hypothesis is that the coefficient is not equal to zero: $H_a: \beta_i \neq 0$. This means the predictor is statistically associated with the log-odds of alcohol involvement, holding all other variables constant.

In logistic regression, the Wald Statistic is used in hypothesis testing. The significance of each coefficient is evaluated using the Wald statistic. The Wald Statistic is defined in terms of:

$$\frac{\hat{\beta}_i}{\sigma_{\hat{\beta}_i}} = Z$$

Where the numerator is the predicted coefficient, and the denominator is the standard error. Under the null hypothesis, the Wald statistic follows a normal distribution. The p-value for each term may be obtained using the standard normal z tables.

In logistic regression, rather than looking at the estimated β coefficients, which represent the log-odds for a one unit increase in the predictor x_k and aren't that intuitive for understanding their influence on a binary variable, most statisticians prefer interpreting odds ratios, which are calculated by exponentiating the coefficients (e^{β_k}). The odds ratio describes the multiplicative change in the odds of the dependent variable being = 1 for a one-unit increase in x_k . In simpler terms, if the coefficient of x is 1.2, the odds ratio equals $e^{1.2} = 3.32$, meaning a one-unit increase

in x is associated with the odds of the DV = 1 being 3.32 times higher, holding all other variables constant.

2.3 Assessing Model Fit

Assessing model fit is done differently for logistic regression than it is for OLS regression. You may calculate the R^2 for logistic regression, but it is not as useful as a metric since it has a different interpretation. In OLS, R^2 is measured as the proportion of variance in y explained by the model, but in logistic regression, your model doesn't seek to capture variance with least squares, it seeks to capture maximum likelihood. While you can calculate pseudo R^2 values, they mean different things than a traditional R^2 and aren't quite comparable.

Instead, to compare models, the Akaike Information Criterion is more effective. AIC is a metric that considers goodness of fit against model complexity (the number of parameters), where lower AIC values mean the model has a better fit.

In logistic regression, the fitted value of $\hat{y}$ is not a prediction of 0 or 1, instead it is an estimated probability that $Y = 1$ for an observation. The model produces the $\hat{y}$ (or $P(Y = 1)$) for each observation by passing the predictors through the logistic function:

$$P(Y = 1) = \hat{y}_i = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \dots + \hat{\beta}_3 x_{3i}}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \dots + \hat{\beta}_3 x_{3i}}}$$

The fitted value of $\hat{y}$ produced is always between 0 and 1, and to make a binary classification (predicts $Y = 0$ or $Y = 1$), you must select a cut-off value. If $\hat{y} > \text{cut-off}$ (often = 0.5), then predict $Y = 1$, otherwise predict $Y = 0$. You should try using different cut-offs instead of arbitrarily calling 0.5 as the cutoff. For instance, lowering the cutoff value to 0.3 may catch more true positives (where the model predicts $Y = 1$ when it is actually $Y = 1$), but may also predict many more false positives (where the model predicts $Y = 1$ when it is actually $Y = 0$). On the flipside, raising the cutoff to 0.7 may predict fewer false positives, but may also miss many true positives. This is the trade-off of selecting a cut-off value: you will either sacrifice sensitivity rates (true positive detection), or specificity rates (true negative detection). This has major implications in the medical field:

Imagine a test for an infectious disease. If the cut-off for detection is low, then the test will be highly likely to catch most positive cases, but will result in many false positives. On the other hand, if the cut-off is high, false-positives will be avoided, but at the cost of missing actually positive cases.

When choosing a cutoff, you either prioritize lowering your false negative rate, or your false positive rate. False positives and negatives can be visualized in a confusion matrix:

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	<i>True Positives (TPs)</i>	<i>False Positives (FPs)</i>
Predicted Negative (0)	<i>False Negatives (FNs)</i>	<i>True Negatives (TNs)</i>

Specificity measures how well the model identifies negatives (where $Y = 0$). Of all the observations that are actually $Y = 0$, what proportion did the model correctly identify? A higher value is better, and produces fewer false positives.

- $\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$

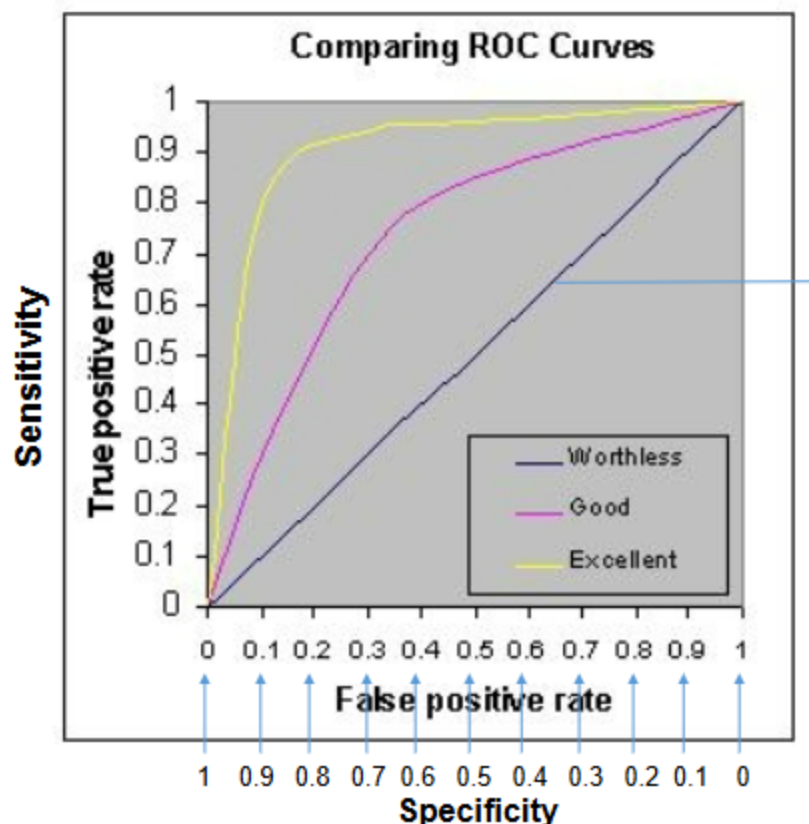
Sensitivity measures how well the model identifies positives (where $Y = 1$). Of all the observations that are actually $Y = 1$, what proportion did the model correctly identify? A higher value is better, and produces fewer false negatives.

- $\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$

The Misclassification Rate is the proportion of observations the model got wrong. A lower value is better, but low values don't always mean a model is useful. For instance, a case where the observation is almost always $Y = 1$ may have a low misclassification rate, despite the model just predicting the majority.

- $\text{Misclassification Rate} = (\text{FP} + \text{FN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

One way of selecting the optimal cut-off value is by using an ROC curve. The Receiver Operating Characteristic (ROC) curve plots sensitivity (true positive rate) on the y-axis and 1 - specificity (false positive rate) on the x-axis. The plot stretches across all possible cut-off values (0 to 1) and each point on the curve corresponds to a different threshold.



A straight ROC curve with a slope of 1/1 means the model predicts no better than randomly guessing, and an excellent ROC curve has a far greater area under the curve (AUC) than above it. In this report, we will weigh specificity and sensitivity equally, so the best cut-off value optimizes sensitivity and specificity, which on the curve, is at the closest point to the top-left of the graph where the true positive rate = 1 and the false positive rate = 0. In other words, you minimize the squared euclidean distance to 0,1 on the graph. Another method is to use the Youden's index, where you maximize distance from the diagonal ($\text{Sensitivity} + \text{Specificity} - 1$). In a case where you want to give more weight to specificity or sensitivity (i.e. you may want a higher sensitivity when detecting serious diseases), you may select a point slightly further from 1,0.

By calculating the area under the curve (AUC), which is always between 0 and 1, you get a value that describes the model's predictive ability. A perfect 1.0 AUC (which would be incredibly difficult to approach) means the model predicts with perfect accuracy, and never produces a false positive or false negative. The following table presents the general guidelines for assessing AUC values:

AUC Range	Model Quality
0.90 - 1.00	Excellent

0.80 - 0.90	Good
0.70 - 0.80	Fair
0.60 - 0.70	Poor
0.50 - 0.60	Fail

2.3 Assumptions of Logistic Regression

Logistic regression lacks some of the key OLS assumptions, and introduces some of its own. Whereas OLS regression requires normally distributed residuals, homoscedasticity, and a linear relationship between predictors and the DV, logistic regression does not. For one, logistic regression predicts for a binary DV ($Y = 0, 1$), so normal residuals are nonsensical. As for homoscedasticity, logistic regression does expect constant variance of residuals across values of the predictors since the variance of an outcome depends directly on the predicted probability. Where OLS requires a linear relationship between predictors and the DV, logistic regression predicts a binary variable, and thus a linear relationship is not possible. Instead, logistic regression assumes the log odds of $Y = 1$ must be linearly related to each of the predictors.

Logistic regression does retain some of the core assumptions of OLS. Observations in logistic regression must be independent and predictors must not be severely collinear. OLS and Logistic regression requires large enough samples to estimate coefficients, but logistic regression requires much greater samples (at least 50 per predictor) to regress.

2.3 Exploratory Analyses for Logistic Regression

Before fitting a logistic regression model, one should explore the relationship between each predictor and the binary dependent variable. By doing so, it gives an early sense of which predictors could be the most useful, and find any potential issues with variables.

One exploratory analysis method is to run a cross-tabulation between each dependent variable and the predictors to see whether there is an association between the two variables. The table shows the distribution of the outcome of the dependent variable across values of the predictor. The table helps to give an immediate visual sense of whether the proportion of $Y = 1$ differs

across groups. The appropriate statistical test for examining the association between two categorical variables is the Chi-Square test of independence. In the context of this report, the hypotheses for the test are:

- H_0 : The proportion of fatalities for crashes that involve drunk drivers is the same as the proportion of fatalities for crashes that don't involve drunk drivers
- H_a : the proportion of fatalities for crashes that involve drunk drivers is different from the proportion of fatalities for crashes that don't involve drunk drivers.

When the Chi-Square test is run, if the p-value is below the significance level (0.05), the null hypothesis is rejected, suggesting that the predictor and outcome are associated and that the predictor may be useful for inclusion in the logistic regression model.

Another preliminary step we can take is to compare the means of continuous predictors for both values of the dependent variable. We can group the means of the values of predictors where $Y = 1$ and where $Y = 0$, and if the means of a predictor differ substantially between groups, the predictor may help discriminate between the two outcomes. The appropriate test for this comparison is the independent samples t-test. The test assesses whether the difference in means between groups is statistically significant. In the context of this report, the independent samples t-test will examine whether there are significant differences in the mean values of PCTBACHMOR and MEDHHINC between crashes that involved alcohol and those that did not. The hypotheses for our test are:

- H_0 : average values of the variable PCTBACHMOR(or MEDHHINC) are the same for crashes that involve drunk drivers and crashes that don't.
- H_a : average values of the variable PCTBACHMOR(or MEDHHINC) are different for crashes that involve drunk drivers and crashes that don't.

3. Results

3.1 Exploratory analysis

The dependent variable was computed using R. The distribution of the dependent variable, DRINKING_D, shows that 2,485 of the 43,364 crashes involved a drinking driver, while 40,879 did not. This means that alcohol-related crashes accounted for 5.73% of all crashes in the sample, compared with 94.27% that did not involve alcohol. These results indicate that crashes involving drunk driving were relatively rare in the dataset.

Distribution of crashes by alcohol involvement			
DRINKING_D	Count	Proportion	Percent
0	40879	0.9427	94.27
1	2485	0.0573	5.73

In addition, we examine cross-tabulations between the dependent variable and the binary predictor variables. Table 1 summarizes the results:

	Non-Alcohol Involved		Alcohol Involved		Total	chi2 stat	df	chi2 p-value
	N	%	N	%				
FATAL_OR_M	1181	2.889	188	7.565	1369	167.562	1	0.000
OVERTURNED	612	1.497	110	4.427	722	122.788	1	0.000
CELL_PHONE	426	1.042	28	1.127	454	0.162	1	0.687
SPEEDING	1261	3.085	260	10.463	1521	376.781	1	0.000
AGGRESSIVE	18522	45.309	916	36.861	19438	67.602	1	0.000
DRIVER1617	674	1.649	12	0.483	686	20.452	1	0.000
DRIVER65PLUS	4237	10.365	119	4.789	4356	80.605	1	0.000

Table 1: Alcohol involvement binary predictors

For each binary predictor, Table 1 reports the chi-square statistic, degrees of freedom, and p-value. The results show that all binary predictors except for CELL_PHONE are significantly associated with DRINKING_D. With this information, for all binary predictors except CELL_PHONE, the chi-square tests provide evidence of a statistically significant association with DRINKING_D. For CELL_PHONE, the p-value is 0.687, so we fail to reject the null hypothesis. The predictor for cell phone use was not statistically significant ($p = 0.687$, which is > 0.05), so we fail to reject the null hypothesis for this variable.

Additionally, we examine whether the means of the two continuous predictors differ at different levels of the dependent variable. Table 2 summarizes the results:

	Non-Alcohol Involved		Alcohol Involved		t-value	df	p-value
	Mean	SD	Mean	SD			
PCTBACHMOR	16.57	18.214	16.612	18.721	-0.108	2777.542	0.914
MEDHHINC	31483.06	16930.102	31998.753	17810.497	-1.405	2763.870	0.160

Table 2: Alcohol-Involved Continuous Predictors

Based on the results in Table 2, we fail to reject the null hypothesis for both continuous predictors. Here, the null hypothesis is that the mean value of each predictor is the same for alcohol-involved and non-alcohol-involved crashes. In other words, the t-tests examine whether PCTBACHMOR and MEDHHINC differ significantly by crash type. Since both p-values are greater than 0.05, there is not enough evidence to conclude that either PCTBACHMOR or MEDHHINC has a statistically significant difference in mean values between alcohol-involved

and non-alcohol-involved crashes. This suggests that the mean values of PCTBACHMOR and MEDHHINC are not statistically significantly different between alcohol-involved and non-alcohol-involved crashes in the bivariate t-tests.

3.2 Assumption check

Pearson correlations are examined between all predictors, both binary and continuous.

Pearson correlation matrix for predictor variables									
	FATAL_OR_M	OVERTURNED	CELL_PHONE	SPEEDING	AGGRESSIVE	DRIVER1617	DRIVER65PLUS	PCTBACHMOR	MEDHHINC
FATAL_OR_M	1.000	0.033	0.002	0.082	-0.011	-0.003	-0.013	-0.015	-0.018
OVERTURNED	0.033	1.000	-0.001	0.059	0.016	0.004	-0.020	0.009	0.028
CELL_PHONE	0.002	-0.001	1.000	-0.004	-0.026	0.001	-0.003	-0.001	0.002
SPEEDING	0.082	0.059	-0.004	1.000	0.212	0.016	-0.033	-0.001	0.012
AGGRESSIVE	-0.011	0.016	-0.026	0.212	1.000	0.028	0.015	0.027	0.043
DRIVER1617	-0.003	0.004	0.001	0.016	0.028	1.000	-0.021	-0.003	0.023
DRIVER65PLUS	-0.013	-0.020	-0.003	-0.033	0.015	-0.021	1.000	0.026	0.050
PCTBACHMOR	-0.015	0.009	-0.001	-0.001	0.027	-0.003	0.026	1.000	0.478
MEDHHINC	-0.018	0.028	0.002	0.012	0.043	0.023	0.050	0.478	1.000

Table 3: Pearson Correlation Matrix for Predictor Variables

One limitation is that Pearson correlations are not ideal for measuring associations among binary variables because binary variables do not have continuous, normally distributed values. For two binary variables, the Pearson correlation is equivalent to the phi coefficient, but it still gives only a simple pairwise association and does not capture more complex multivariable relationships.

From the Pearson correlation matrix, the largest correlation is 0.478 between PCTBACHMOR and MEDHHINC. This suggests that there is no evidence of severe multicollinearity among the predictors. We also note that the binary structure of DRINKING_D makes logistic regression appropriate as a modeling framework. Although it is not directly tested here, we assume that the log-odds, or logit, of the outcome variable is linearly related to the predictors. Finally, based on the study design, the observations are assumed to be independent. However, because crashes are spatial events, there may be some spatial or clustered dependence that is not captured by this model.

3.3 Logistic Regression Results

Call:

```
glm(formula = DRINKING_D ~ FATAL_OR_M + OVERTURNED + CELL_PHONE +
    SPEEDING + AGGRESSIVE + DRIVER1617 + DRIVER65PLUS + PCTBACHMOR +
    MEDHHINC, family = "binomial", data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.732506616	0.045875659	-59.563	< 0.0000000000000002 ***
FATAL_OR_M	0.814013802	0.083806924	9.713	< 0.0000000000000002 ***
OVERTURNED	0.928921376	0.109166324	8.509	< 0.0000000000000002 ***
CELL_PHONE	0.029550085	0.197777821	0.149	0.8812
SPEEDING	1.538975665	0.080545894	19.107	< 0.0000000000000002 ***
AGGRESSIVE	-0.596915946	0.047779238	-12.493	< 0.0000000000000002 ***
DRIVER1617	-1.280295964	0.293147168	-4.367	0.000012572447127933 ***
DRIVER65PLUS	-0.774664640	0.095858315	-8.081	0.000000000000000641 ***
PCTBACHMOR	-0.000370634	0.001296387	-0.286	0.7750

```

MEDHHINC      0.000002804  0.000001341  2.091      0.0365 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for binomial family taken to be 1)
Null deviance: 19036  on 43363  degrees of freedom
Residual deviance: 18340  on 43354  degrees of freedom
AIC: 18360
Number of Fisher Scoring iterations: 6

```

	Estimate	Std. Error	z value	Pr(> z)	OR	2.5 %	97.5 %
(Intercept)	-2.733	0.046	-59.563	0.000	0.065	0.059	0.071
FATAL_OR_M	0.814	0.084	9.713	0.000	2.257	1.910	2.653
OVERTURNED	0.929	0.109	8.509	0.000	2.532	2.035	3.122
CELL_PHONE	0.030	0.198	0.149	0.881	1.030	0.684	1.488
SPEEDING	1.539	0.081	19.107	0.000	4.660	3.974	5.450
AGGRESSIVE	-0.597	0.048	-12.493	0.000	0.551	0.501	0.604
DRIVER1617	-1.280	0.293	-4.367	0.000	0.278	0.148	0.471
DRIVER65PLUS	-0.775	0.096	-8.081	0.000	0.461	0.380	0.553
PCTBACHMOR	0.000	0.001	-0.286	0.775	1.000	0.997	1.002
MEDHHINC	0.000	0.000	2.091	0.036	1.000	1.000	1.000

Table 4: Logistic Regression Results

Table 4 presents the full logistic regression model output predicting whether a crash involved alcohol. Several predictors were statistically significant. Holding all other variables constant, crashes involving speeding had much higher odds of alcohol involvement (OR = 4.66, $p < 0.001$), making speeding the strongest positive predictor in the model. Similarly, crashes that involved an overturned vehicle were associated with greater odds of alcohol involvement (OR = 2.53, $p < 0.001$), and crashes resulting in a fatality or major injury also had higher odds of involving alcohol (OR = 2.26, $p < 0.001$). In substantive terms, this means that the odds of a crash being alcohol-related were 366% higher when speeding was involved, 153% higher when a vehicle overturned, and 126% higher when the crash resulted in a fatality or major injury, compared with crashes where those conditions were not present.

Several variables were associated with lower odds of alcohol involvement. Aggressive driving was negatively associated with alcohol-related crashes (OR = 0.55, $p < 0.001$), indicating that the odds of alcohol involvement were about 45% lower when aggressive driving was present. Crashes involving at least one driver aged 16 or 17 were also less likely to involve alcohol (OR = 0.28, $p < 0.001$), corresponding to about 72% lower odds, while crashes involving a driver aged 65 or older had about 54% lower odds of alcohol involvement (OR = 0.46, $p < 0.001$). In contrast, cell phone use was not a significant predictor of alcohol involvement (OR = 1.03, $p = 0.881$). Among the neighborhood-level variables, PCTBACHMOR was not statistically significant (OR = 1.00, $p = 0.775$), while MEDHHINC was statistically significant ($p = 0.036$) but had an odds ratio essentially equal to 1.00, indicating only a trivial effect size. Because median household income is measured in dollars, the odds ratio for a one-dollar increase is extremely small in practical terms, so this result is likely statistically significant mainly because of the very large sample size rather than because it represents a meaningful substantive effect.

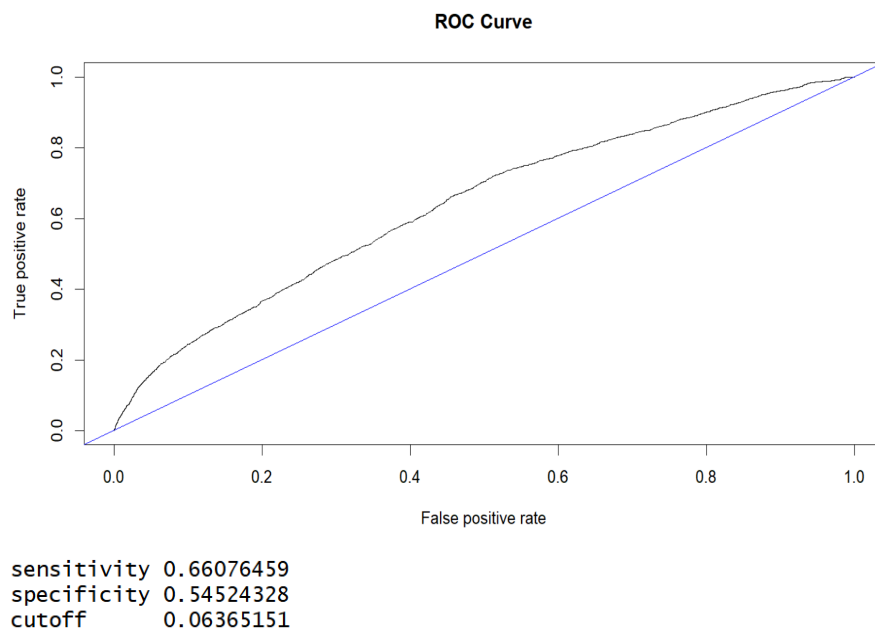
3.4 Model Classification Performance

Table 5 reports the sensitivity, specificity, and misclassification rate for each probability cut-off. As shown in the table, the cut-off of 0.02 produces the highest misclassification rate, at 0.889. At this threshold, the model classifies nearly all crashes as alcohol-involved, which leads to very high sensitivity but very low specificity. In contrast, the cut-off of 0.50 yields the lowest misclassification rate, at 0.057. Overall, the results show that the misclassification rate declines as the cut-off increases. However, the 0.50 cutoff is not substantively useful for identifying alcohol-involved crashes because sensitivity is only 0.002. This means the model almost never classifies a crash as alcohol-involved. The low misclassification rate is mainly driven by the fact that most crashes are non-alcohol-involved. Therefore, the ROC-based cutoff is more informative when the goal is to balance true positive and false positive classification.

	Sensitivity	Specificity	Misclassification Rate
Cutoff=0.02	0.984	0.058	0.889
Cutoff=0.03	0.981	0.064	0.884
Cutoff=0.05	0.735	0.469	0.516
Cutoff=0.07	0.221	0.914	0.126
Cutoff=0.08	0.185	0.939	0.105
Cutoff=0.09	0.168	0.946	0.099
Cutoff=0.1	0.164	0.948	0.097
Cutoff=0.15	0.104	0.972	0.078
Cutoff=0.2	0.023	0.995	0.060
Cutoff=0.5	0.002	1.000	0.057

Table 5: Misclassification rates based on different probability cutoffs

ROC curve and AUC



This graph presents the Receiver Operating Characteristic (ROC) curve for Model 1, which shows the tradeoff between sensitivity (the true positive rate) and $1 - \text{specificity}$ (the false positive rate) across all possible probability cut-offs. The curve lies above the 45-degree reference line, indicating that the model performs better than random classification in distinguishing alcohol-involved crashes from non-alcohol-related crashes. To select a single operating threshold from the ROC space, the cut-off was chosen by minimizing the squared Euclidean distance to the upper-left corner, (0, 1), which represents perfect classification. Using this criterion, the optimal cut-off was 0.0637, which produced a sensitivity of 0.661 and a specificity of 0.545. This means that, at that threshold, the model correctly identifies about 66.1% of alcohol-involved crashes and about 54.5% of crashes that did not involve alcohol. Thus, the ROC-based cut-off provides a more balanced tradeoff between detecting alcohol-involved crashes and limiting false positives than a higher threshold would.

The area under the ROC curve (AUC) for Model 1 is 0.6399, indicating modest discriminatory power. Conceptually, the AUC can be interpreted as the probability that the model assigns a higher predicted probability to a randomly selected alcohol-involved crash than to a randomly selected non-alcohol-involved crash. An AUC of 0.6399 means that, about 64% of the time, the model assigns a higher predicted probability to a randomly selected alcohol-involved crash than to a randomly selected non-alcohol-involved crash, but it does not distinguish particularly well between the two outcome groups across thresholds. This ROC-based optimal cut-off also differs substantially from the cut-off selected in Table 5 using the minimum misclassification rate criterion. In Table 5, the misclassification rate was lowest at a cut-off of 0.50, but that threshold yielded extremely low sensitivity (0.002) and specificity (1.000), indicating that the model classified almost all crashes as non-alcohol-involved. By contrast, the ROC-based threshold of

about 0.064 weights sensitivity and specificity more evenly, leading to much better identification of alcohol-involved crashes even though it necessarily lowers specificity and increases false positives relative to the 0.50 cut-off.

```
Call:
glm(formula = DRINKING_D ~ FATAL_OR_M + OVERTURNED + CELL_PHONE +
     SPEEDING + AGGRESSIVE + DRIVER1617 + DRIVER65PLUS, family = "binomial",
     data = data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.65190	0.02753	-96.324	< 0.0000000000000002 ***
FATAL_OR_M	0.80932	0.08376	9.662	< 0.0000000000000002 ***
OVERTURNED	0.93978	0.10903	8.619	< 0.0000000000000002 ***
CELL_PHONE	0.03107	0.19777	0.157	0.875
SPEEDING	1.54032	0.08053	19.128	< 0.0000000000000002 ***
AGGRESSIVE	-0.59365	0.04775	-12.433	< 0.0000000000000002 ***
DRIVER1617	-1.27158	0.29311	-4.338	0.00001436374143265 ***
DRIVER65PLUS	-0.76646	0.09576	-8.004	0.00000000000000121 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 19036 on 43363 degrees of freedom
Residual deviance: 18344 on 43356 degrees of freedom
AIC: 18360

Number of Fisher Scoring iterations: 6

	Estimate	Std. Error	z value	Pr(> z)	OR	2.5 %	97.5 %
(Intercept)	-2.652	0.028	-96.324	0.000	0.071	0.067	0.074
FATAL_OR_M	0.809	0.084	9.662	0.000	2.246	1.901	2.640
OVERTURNED	0.940	0.109	8.619	0.000	2.559	2.057	3.156
CELL_PHONE	0.031	0.198	0.157	0.875	1.032	0.685	1.491
SPEEDING	1.540	0.081	19.128	0.000	4.666	3.980	5.457
AGGRESSIVE	-0.594	0.048	-12.433	0.000	0.552	0.503	0.606
DRIVER1617	-1.272	0.293	-4.338	0.000	0.280	0.149	0.475
DRIVER65PLUS	-0.766	0.096	-8.004	0.000	0.465	0.383	0.558

	df <dbl>	AIC <dbl>
model1	10	18359.63
model2	8	18360.47

Table 6: Reduced Model and Model Comparison

The reduced logistic regression model, which includes only the binary predictors, is highly consistent with the full model that includes the two continuous variables. In the reduced model, the same overall pattern of statistical significance remains, and the estimated effects are very similar to those in the full model. FATAL OR M, OVERTURNED, SPEEDING, AGGRESSIVE, DRIVER1617, and DRIVER65PLUS are statistically significant with $p < 0.001$ and comparable odds ratios, while CELL_PHONE remains not statistically significant ($OR = 1.03$, $p = 0.875$). Overall, among the predictors included in both models, the pattern of statistical significance is unchanged, and the estimated odds ratios are very similar across the two specifications. This suggests that removing the two continuous neighborhood-level variables does not significantly change the estimated relationships between the binary crash characteristics and alcohol involvement.

The two models were compared using the Akaike Information Criterion (AIC), where lower values indicate a better fit. The full model has a slightly lower AIC than the reduced model (18359.63 versus 18360.47), but the difference is less than 1. This provides little evidence that the full model is meaningfully better. Because the reduced model is simpler and produces nearly identical results for the binary predictors, the reduced model may be preferred for parsimony, while the full model is technically slightly better by AIC.

4. Discussion

In this paper, we used logistic regression to examine which crash-level and neighborhood-level variables were associated with whether a crash in Philadelphia involved a drinking driver. The strongest positive predictor was SPEEDING: holding the other variables constant, crashes involving speeding had about 4.66 times the odds of alcohol involvement compared with crashes that did not involve speeding. OVERTURNED and FATAL_OR_M were also strong positive predictors, with odds ratios of about 2.53 and 2.26, respectively. In contrast, AGGRESSIVE, DRIVER1617, and DRIVER65PLUS were negatively associated with alcohol involvement. CELL_PHONE and PCTBACHMOR were not statistically significant predictors in the full model. MEDHHINC was statistically significant, but its effect size was very small when measured in one-dollar units; even when rescaled, the substantive effect remained modest. The reduced model with only binary predictors produced nearly identical results for the shared predictors, and although the full model had a slightly lower AIC, the difference was very small. Therefore, the reduced model may be preferable for parsimony, while the full model is only marginally better by AIC.

Some findings were consistent with expectations, while others were more surprising. The negative relationship for DRIVER1617 was expected, since drivers aged 16 or 17 may have less access to alcohol and fewer opportunities to drive after drinking. The positive relationships for SPEEDING, OVERTURNED, and FATAL_OR_M also make substantive sense because alcohol involvement may be associated with more severe or risky crash circumstances. However, the negative coefficient for AGGRESSIVE is less intuitive and may reflect how crash reports classify aggressive driving separately from alcohol-related behaviors, rather than showing that aggressive driving is truly protective. The neighborhood-level variables were weaker than expected: PCTBACHMOR was not significant, and MEDHHINC had only a very small practical effect. This suggests that the immediate crash characteristics were more useful predictors of alcohol involvement than the Census block group socioeconomic variables included in this model.

Logistic regression is appropriate for this analysis because the dependent variable is binary and the number of alcohol-involved crashes is large enough for standard logistic regression. Although alcohol-involved crashes are relatively uncommon, they are not extremely rare in this dataset: 2,485 of 43,364 crashes, or 5.73%, had DRINKING_D = 1. Because there are thousands

of events and only nine predictors in the full model, rare-events logistic regression methods, such as those discussed by Paul Allison, are not necessary here. The more important issue is class imbalance. Since most crashes did not involve alcohol, the 0.50 cutoff produced the lowest misclassification rate mainly by classifying almost every crash as non-alcohol-involved. For this reason, the ROC-based cutoff of about 0.064 is more useful if the goal is to identify alcohol-involved crashes, even though it increases false positives.

This analysis also has several limitations. First, the model should be interpreted as associational, not causal. Variables such as speeding, fatal injury, and overturned vehicles describe characteristics of the crash, but the model cannot prove that these factors caused alcohol involvement. Second, the data do not include several likely important predictors, such as time of day, day of week, road type, speed limit, proximity to bars or nightlife, weather, traffic volume, or enforcement patterns. Third, crashes are spatial events and may not be fully independent; crashes in the same block group or along the same corridors may share unobserved roadway and neighborhood conditions. Finally, the model's AUC of about 0.64 indicates only modest discriminatory power, so the model is useful for identifying broad associations but should not be treated as a highly accurate prediction tool for classifying individual crashes.